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Prospective cohort study of regenerative potential of non vital immature permanent maxillary central incisors using platelet rich fibrin scaffold

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Pulp necrosis as a result of trauma or other insults interrupts the process of root maturation and apical closure in young permanent teeth¹. A twelve-year review of literature states that 25% of school children and 33% of adults experience trauma to their permanent dentition with the majority of these occurring before the age of 19 years². The treatment of young immature permanent teeth with pulp necrosis is therefore challenging to dental practitioners³. A paradigm shift towards regenerative endodontics procedures (REP) facilitated continued root maturation and apical closure. REP is defined “as a biological procedure that predictably replaces damaged, diseased or missing structures including dentin and root structures as well as cells of the pulp-dentin complex (PDC), with viable tissues preferably of the same origin, that restore the normal physiologic functions of the PDC”⁴.

Tissue engineering and tissue regeneration form the backbone of REP; the former utilizes specific stem cells, three-dimensional scaffolds and growth factors to regenerate PDC, the latter involves disinfection and revascularization of the PDC^{4,5}. do Couto et al. stated that the disinfection of the immature teeth in REP can be judiciously achieved with the use of triple antibiotic paste (TAP)⁶. Although the current clinical recommendations by the American Association of Endodontists (AAE) for REP suggests the use of TAP or calcium hydroxide, when comparatively analysing, TAP and double antibiotic paste showed increased antimicrobial activity compared to calcium hydroxide⁷.

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A scaffold is a physicochemical three-dimensional structure facilitating cellular organization and vascularisation. Natural scaffolds such as platelet-rich plasma (PRP) and platelet-rich fibrin (PRF) are advantageous over synthetic polymers and collagen in terms of cost, inflammatory reactions, immune responses and cytotoxicity⁸.

According to the currently available data, revascularization through blood clot technique and scaffold implantation remains the techniques that have been clinically attempted. Although revascularization using a blood clot was effective, the clot formed was unpredictable to serve as a scaffold. On the other hand, scaffold implantation with platelet concentrates delivers an enhanced concoction of growth factors and offers mechanical sustenance⁹. Visser et al. reported that the platelet concentrates have increased concentration of growth factors and increased cellular proliferation when compared to the blood clot¹⁰.

Platelet-rich fibrin is a second-generation platelet concentrate. Ehrenfest et al. classified platelet concentrate into four main categories namely, pure PRP (P-PRP), leukocyte-rich PRP (L-PRP), pure PRF (P-PRF) and leukocyte-rich PRF (L-PRF). The leukocyte-rich PRF includes the Choukron's PRF, advanced PRF (A-PRF) and injectable PRF (I-PRF)¹¹. PRF is an autologous tri-molecular or equilateral fibrin branched junctional matrix which is expedient over PRP in terms of ease of preparation and use of anticoagulants^{12,13}. According to Huang et al., PRF can stimulate pulp cell proliferation, enhance osteoprotegerin protein expression and induce alkaline phosphatase activity⁵.

According to data analysis on clinical protocols in REP, the overall percentage of studies in which intracanal bleeding or PRP/PRF was used is only 13% with insufficient evidence using a large sample size available in the literature¹⁴.

Hence the aim of this in vivo prospective cohort study was to evaluate the regenerative potential of scaffold implantation technique with PRF in non-vital immature permanent maxillary central incisors with a follow-up period of 2 years. In accordance with the AAE guidelines (8.5.2014), the primary and secondary goals were set as follows:

Primary: Elimination of the symptoms of pain /swelling (before scaffold implantation procedure) with evidence of bony healing and resolution of apical radiolucency (observed 6–12 months after treatment).

Secondary: Increase in root length and root dentin thickness (observed at 12–24 months after treatment) and decrease in apical diameter.

Tertiary: Positive pulp vitality response (which if achieved, could indicate a more organized vital pulp tissue).

Methods

Patient recruitment. Nineteen patients (4 patients with 2 teeth each) in the age group of 9–25 years with immature, non-vital permanent maxillary central incisors ($n = 23$) with/without signs and/or symptoms of peri-apical pathosis (PAI index-up to scale 2—small changes in bone structure) and open apex (apical diameter greater than 1.1 mm) were included in this study. The study protocol was presented to the Institutional Review Board and approval was obtained based on the regulation of the Ethical Committee of the Faculty of Dentistry, SRM Institute of Science and Technology (SRMDC/IRB/2014/MDS/No.304) and registered in Clinical Trial Registry of India (CTRI/2017/10/010,246) (30/10/2017). Patients were explained about the treatment protocol and informed consent was obtained from the patient and his/her parent (if the patient was less than 18 years of age). Detailed medical and dental histories, and clinical examination with pulp sensibility testing and radiographic examination of the involved teeth were carried out to confirm the diagnosis. Teeth with closed apex, caries lesions, periodontally compromised teeth, resorption, teeth with root fracture, teeth other than maxillary central incisors were excluded. All procedures were carried out in accordance with the Ethical Committee guidelines of SRM Institute of Science and Technology and AAE guidelines for regenerative endodontics (8.5.2014). The flowchart according to the STROBE guidelines is depicted in Fig. 1.

Treatment procedure. *Appointment 1.* Local anesthesia (LA) was administered using 2% lignocaine without vasoconstrictor by infiltration technique. Under rubber dam isolation and aseptic conditions, the access cavity was prepared using sterile No:1 Endo access and Endo 'Z' bur (Dentsply Maillefer, USA). After working length (WL) determination, minimal instrumentation was done and the canal was irrigated using 10 ml of 1% sodium hypochlorite (NaOCl) (Chenchems Reagent, Chenchemicals, Chennai, India) for 2 min with side vented needles placed 1 mm short of WL. The canals were dried using paper points 2 mm short of the WL. 0.02 mg each of ciprofloxacin, metronidazole and amoxicillin was mixed in the ratio of 1:1:1 with 0.5 mL of propylene glycol to obtain a final concentration of 0.12 mg/mL which was placed in the coronal third of the root canal. The pulp chamber was then temporarily restored with Cavit G (3 M ESPE, Germany) and a radiograph was taken. The patient was recalled after 3 weeks.

Appointment 2. Preparation of PRF. 10 mL of whole blood was withdrawn from the median cubital vein of the patient using 10 mL disposable syringe and needle. The blood was collected in a sterile glass test tube and centrifuged (REMI centrifuge, Mumbai, India) at 3300 rpm for 12 min at 400 g force. The resultant membrane was then placed between 2 sterile gauze pieces and squeezed to form a strong fibrin network.

Placement of PRF. 3% mepivacaine without vasoconstrictor (Scanodest, Septodont, Lancaster, UK) was administered. Under aseptic conditions and rubber dam isolation, the temporary restoration was removed and the TAP was flushed out copiously with saline. Canal irrigation was done using 10 mL of 1% NaOCl and a final rinse with 5 mL of 17% EDTA (Chenchems Reagent, Chenchemicals, Chennai, India) solution. The canal was dried with paper points and the thin strands of PRF were cut and placed into the root canal space using Buchanan hand pluggers (Sybron Endo, Glendora, CA) 1 mm beyond the working length and coronally just below the CEJ. MTA (MTA Angelus, Angelus, Brazil) plug of 3 mm was packed coronally over the PRF scaffold. The canal orifice was

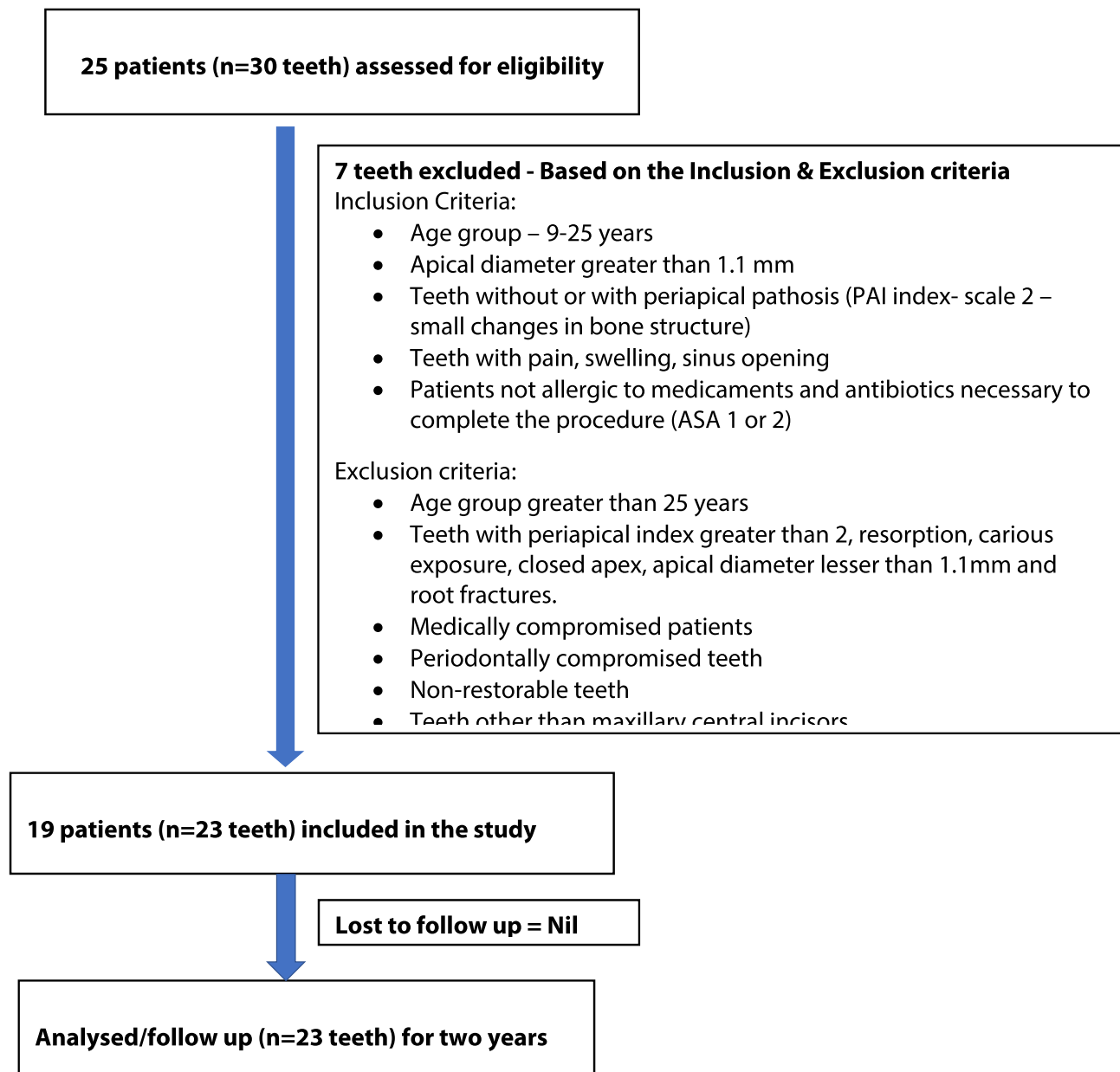


Figure 1. Flowchart according to STROBE guidelines.

sealed with Type II glass ionomer cement (GC, Tokyo, Japan) and the tooth was finally restored using composite resin (Te-Econom Plus, Ivoclar Vivadent, Mumbai, India) and a digital radiograph was taken.

Follow-up analysis. The routine follow-up was carried out using clinical and radiographic assessments at 3, 6, 9, 12, 18, 24, 30, 36, 42, 48, 54, 60, 66, 72, 78, 84, 90, 96, 102, 108, 114, 120, 126, 132, 138, 144, 150, 156, 162, 168, 174, 180, 186, 192, 198, 204, 210, 216, 222, 228, 234, 240, 246, 252, 258, 264, 270, 276, 282, 288, 294, 300, 306, 312, 318, 324, 330, 336, 342, 348, 354, 360, 366, 372, 378, 384, 390, 396, 402, 408, 414, 420, 426, 432, 438, 444, 450, 456, 462, 468, 474, 480, 486, 492, 498, 504, 510, 516, 522, 528, 534, 540, 546, 552, 558, 564, 570, 576, 582, 588, 594, 600, 606, 612, 618, 624, 630, 636, 642, 648, 654, 660, 666, 672, 678, 684, 690, 696, 702, 708, 714, 720, 726, 732, 738, 744, 750, 756, 762, 768, 774, 780, 786, 792, 798, 804, 810, 816, 822, 828, 834, 840, 846, 852, 858, 864, 870, 876, 882, 888, 894, 900, 906, 912, 918, 924, 930, 936, 942, 948, 954, 960, 966, 972, 978, 984, 990, 996, 1002, 1008, 1014, 1020, 1026, 1032, 1038, 1044, 1050, 1056, 1062, 1068, 1074, 1080, 1086, 1092, 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Parameters	Pre-Op	3 Months	6 Months	9 Months	12 Months	15 Months	18 Months	24 Months
Apical Diameter (mm)	0 ^a	0 ^a	0 ^a	0.300 ^a (0.47)	0.83^b (0.38)	1.09^b (0.288)	1.35^b (0.487)	1.83^b (0.388)
Root Length (mm)	0 ^a	0 ^a	0 ^a	0.48 ^a (0.51)	0.91^b (0.28)	1.17^b (0.388)	1.57^b (0.507)	1.91^b (0.417)
Thickness of Mesial Dentinal wall- Apical third (mm)	0 ^a	0 ^a	0 ^a	0.04 ^a (0.20)	0.70^b (0.47)	0.96^b (0.209)	1.00^b (0.000)	1.04^b (0.209)
Thickness of Distal Dentinal wall – Apical third (mm)	0 ^a	0 ^a	0 ^a	0.43 ^a (0.50)	0.87^b (0.34)	1.00^b (0.000)	1.04^b (0.209)	1.04^b (0.209)
Thickness of Mesial Dentinal wall- Middle third (mm)	0 ^a	0 ^a	0 ^a	0 ^a	0.13 ^a (0.34)	0.30 ^a (0.470)	0.48^b (0.511)	0.57^b (0.507)
Thickness of Distal Dentinal wall- Middle third (mm)	0 ^a	0 ^a	0 ^a	0 ^a	0 ^a	0.26 ^a (0.449)	0.30 ^a (0.470)	0.52^b (0.511)

Table 1. Mean (Standard deviation) of difference between Pre- and Post-operative values of various parameters. ^{a,b}Different alphabetical superscript for each time period indicates the statistical significant difference ($p < 0.05$).

done by cold test using ethyl-chloride spray (Roeko Endo-Frost, COLTENE, USA) and heat test using hot water at each time interval.

factors³⁷.

Conclusion

Within the limitations of the present in vivo prospective cohort study, it can be concluded that the use of platelet-rich fibrin as a scaffold has proven to be clinically and radiographically effective in terms of increase in root length, decrease in apical diameter and increase in dentin wall thickness in all subjects when tested over a 2 year follow up period. Thus regenerative endodontics using scaffold implantation technique (PRF) can be a viable alternative treatment option over other techniques for non-vital immature permanent teeth.



Figure 2. Case 1: Tooth no. 11, 21- (A) Preoperative radiograph; (B–D) Radiographs at 6 months, 12 months and 24 months follow-up period.

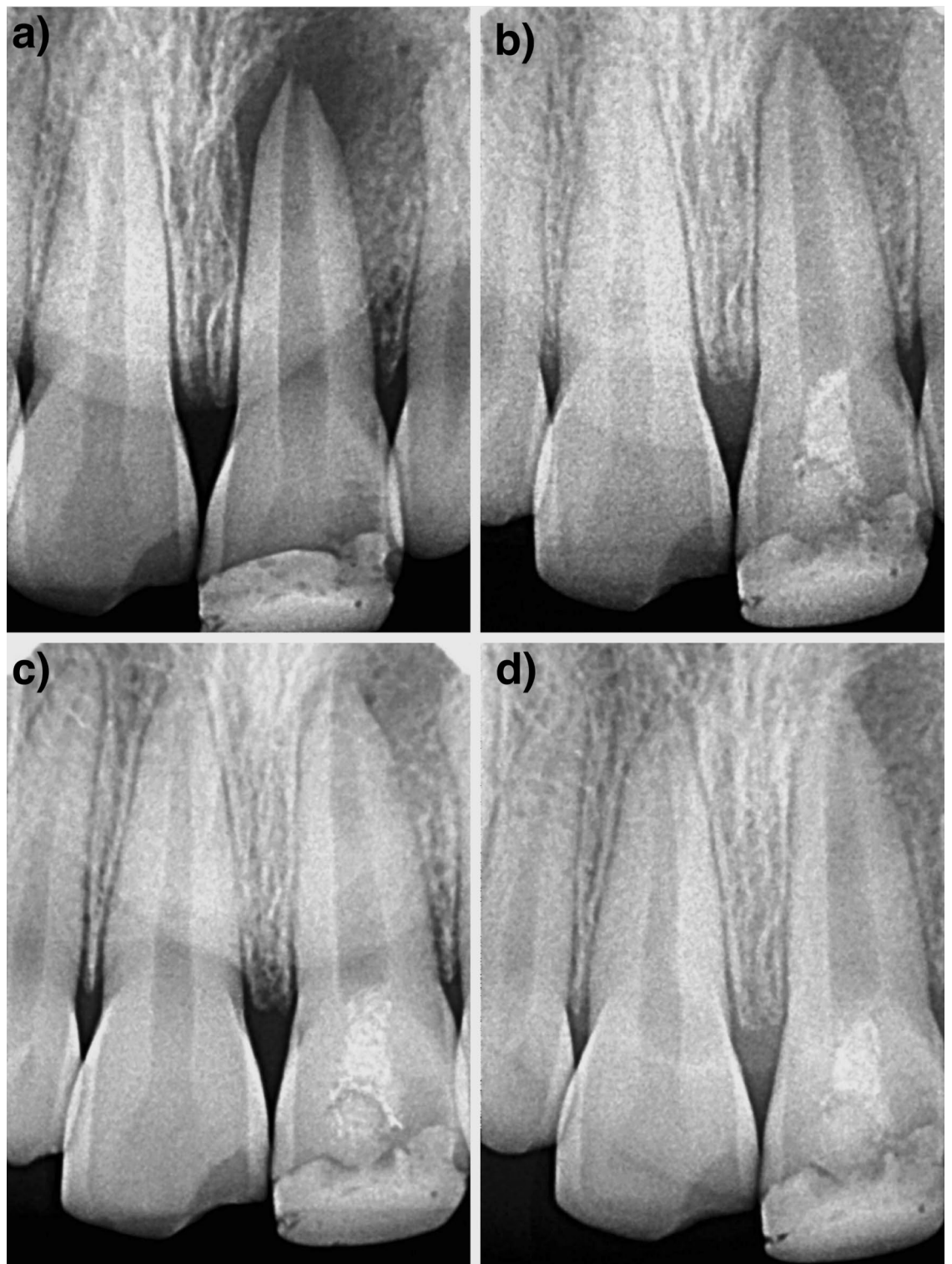


Figure 3. Case 2: Tooth no. 21- (A) Preoperative radiograph; (B–D) Radiographs at 6 months, 12 months and 24 months follow-up period.

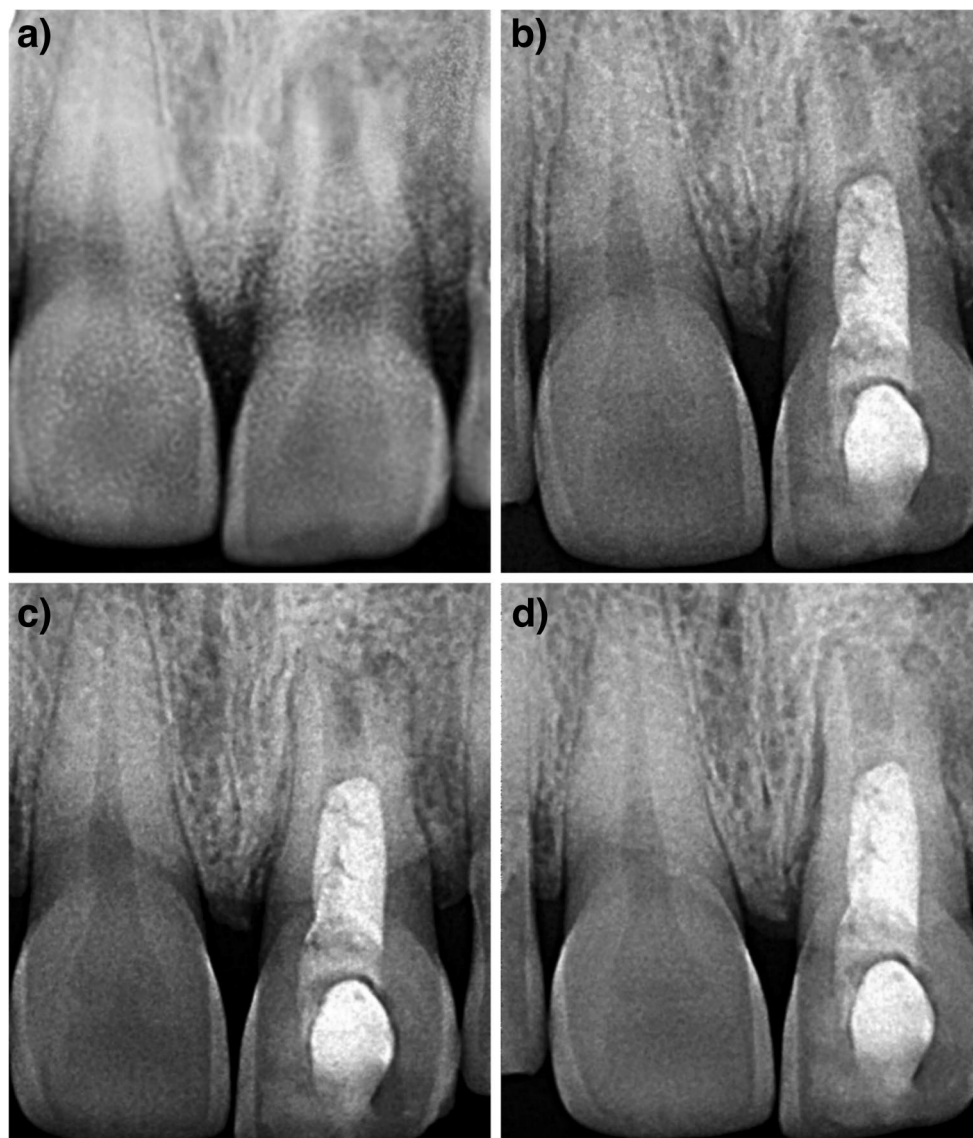


Figure 4. Case 3: Tooth no. 21- (A) Preoperative radiograph; (B–D) Radiographs at 6 months, 12 months and 24 months follow-up period.

Data availability

All data generated or analysed during this study are included in this published article (and its Supplementary Information files).

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Competing interests

The authors declare no competing interests.

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